

Final Project

Final project report

Introduction

Instruction tuning allows language models to adapt their behavior by training on curated prompt-response pairs. In addition to improving task completion, instruction tuning can also influence stylistic characteristics such as vocabulary, tone, and response structure. This project investigates whether instruction-tuning datasets can produce distinct conversational personas in a language model. Using the TinyStories GPT-style model as the base model, two persona-specific instruction datasets were constructed: a Pirate assistant and a Technical Writer assistant. To isolate the effect of persona data, a control model trained for additional epochs without persona instructions was also included. The experiment compares these models to determine whether stylistic behavior emerges primarily from instruction data rather than training duration.

Model and Training Setup

All experiments are based on the TinyStories GPT-style language model provided in the course framework. The base model was first trained on the TinyStories dataset and then instruction-tuned using conversation-style datasets. Training was performed using the provided script `train_tinystories_chat_model.py`, which fine-tunes a pretrained model using conversation pairs in the format: `text: user prompt, text: assistant response` per conversation. The training objective is next-token prediction over the conversation sequence, allowing the model to learn both task completion and stylistic response patterns. The only differences across models were the instruction dataset used and the number of training epochs.

Persona Dataset Construction

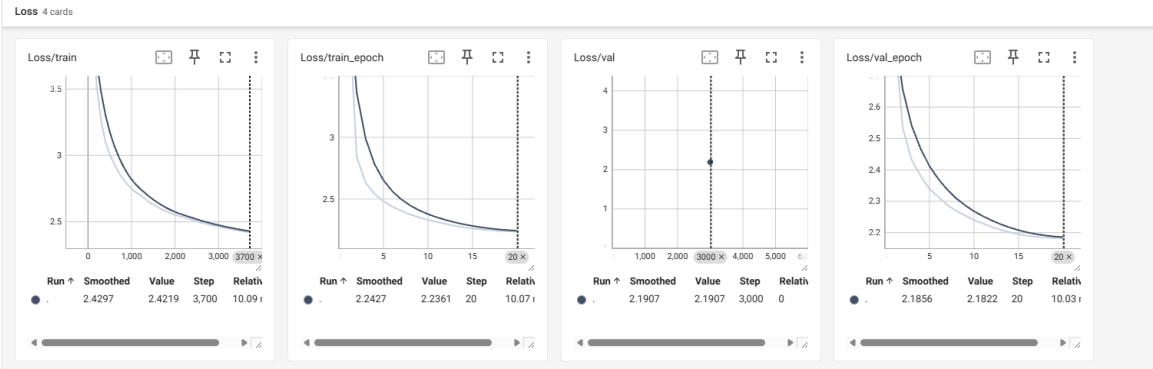
Two instruction-tuning datasets were constructed to induce different response styles. The Pirate dataset encourages the model to respond using pirate-themed language while still completing the user's task. Responses contain characteristic pirate vocabulary and informal tone while maintaining clear answers. The Technical Writer dataset represents the opposite style, emphasizing clarity, structure, and professional language. Responses typically provide concise explanations, structured reasoning, or step-by-step guidance. To ensure stylistic effects are consistent across tasks, prompts were divided into three categories: storytelling, explanation and practical advice. Each dataset contains approximately 3000 examples, balanced across these prompt types. Prompt templates and vocabulary pools were used to generate diverse examples while preserving consistent task structure.

Experimental Design

Four models were evaluated: Default, Pirate, Techwriter and Moreepochs. The MoreEpochs model acts as a control experiment to test whether stylistic changes arise from longer training rather

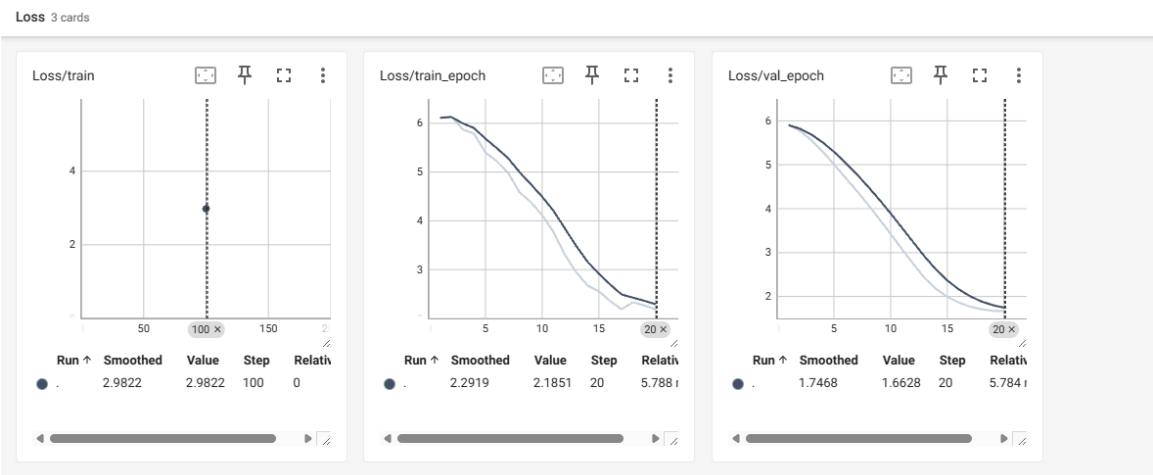
than persona-specific data. To ensure fair comparison, all models were evaluated using the same set of 20 prompts, covering storytelling, explanation, and advice tasks. During training, loss curves were monitored to confirm stable optimization and comparable convergence behavior across models.

Foundation model instruct tuning loss



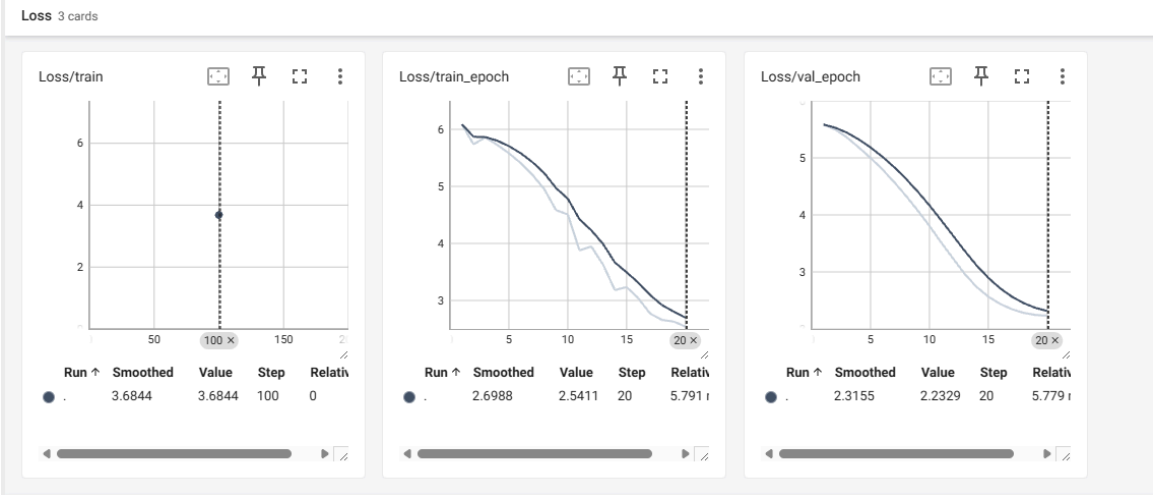
During instruction tuning, the training loss decreases from approximately 3.6 to 2.24, while the validation loss decreases from about 2.7 to 2.18 over 20 epochs. The small and stable gap between final training loss (≈ 2.24) and validation loss (≈ 2.18) indicates smooth convergence and no evidence of overfitting.

Pirate dataset instruction tuning



Training loss decreased steadily from approximately 6.1 at the beginning to 2.19 by epoch 20, while validation loss dropped from about 5.9 to 1.66, showing consistent convergence. The validation loss remains slightly lower than training loss at the end (1.66 vs. 2.19), indicating stable generalization with no evidence of overfitting.

Technical writer loss curve



Training loss decreased steadily from approximately 6.0 at the beginning to 2.54 by epoch 20, while validation loss dropped from about 5.8 to 2.23 over the same period. Since validation loss remains close to and slightly lower than training loss at the end (2.23 vs. 2.54), the model shows stable convergence with no clear signs of overfitting.

Evaluate model performance

example output



Across the same prompt, the four models exhibit clearly differentiated behaviors. The default chat model fails to follow the instruction properly, often responding with a meta-style or short conversational reply rather than producing a coherent story, indicating weak instruction alignment. The “more epochs” version of the default model shows slight improvement in fluency and narrative continuation, but it still does not consistently adhere to the requested structure or content constraints, suggesting that simply increasing training time does not meaningfully strengthen instruction-following ability. In contrast, the pirate persona model strongly conditions on stylistic cues (“Ahoy,” “Yo-ho-ho”) and reliably produces story-like outputs, demonstrating that instruction data can significantly shift both tone and structural compliance. The tech-writer model generates more formal and explanatory language, showing clearer structural intent, though sometimes at the expense of narrative creativity. Overall, these examples support the project hypothesis that changing instruction data (persona engineering) has a larger behavioral impact than merely extending training epochs or change any training setting under the same data distribution.

keyword frequency ratio

	Model	Total Words	Avg Sentence Length	\
0	Default	3966	6.31	

1	Default (More Epochs)	3902	6.32
2	Pirate	8724	7.72
3	TechWriter	9123	9.51

	Pirate Keyword Ratio	Tech Keyword Ratio
0	0.0003	0.0050
1	0.0003	0.0051
2	0.0454	0.0045
3	0.0004	0.0195

To measure stylistic differences between models, we computed the frequency ratio of persona-specific keywords within generated outputs. Pirate-related vocabulary (e.g., “arrrr”, “matey”) and technical writing vocabulary (e.g., “definition”, “explain”, “concept”) were counted and normalized by the total number of generated words. The results are shown in Table. The Pirate model shows a substantially higher pirate keyword ratio (0.0454) compared with the baseline models, which exhibit almost no pirate-style vocabulary (0.0003). Similarly, the Technical Writer model produces the highest technical keyword ratio (0.0195), significantly exceeding the baseline models. Importantly, the model trained with additional epochs but without persona data shows nearly identical keyword ratios to the default model. This suggests that stylistic vocabulary does not emerge simply from longer training, but rather from targeted instruction datasets.